

Three mechanisms and their uses

A mathematical companion to the Courtade–Kumar manuscript

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Companion to edition v11. The full manuscript supplies every proof. The claims below are those of the proposed argument, not a report of journal acceptance or completed formal verification.

1. What the result would let a reader conclude

Take independent fair signs $X = (X_1, \dots, X_n)$. Reverse each sign independently with probability p to obtain Y . Before the noise is drawn, choose a deterministic one-bit compression $f(X)$. The proposed bound is

$$I(f(X); Y) \leq \ln 2 - h(p), \quad h(p) = -p \ln p - (1 - p) \ln(1 - p).$$

A coordinate indicator attains this value. The quantifiers are all finite n , every output mean, and $0 \leq p \leq 1/2$. Uniform inputs and independent symmetric noise are hypotheses, not conveniences that may be dropped.

There is also an explicit structural conclusion. Write $F = 2f - 1$, $\rho = 1 - 2p$, and

$$\delta(f) = \min_{i, \sigma} \mathbb{P}\{F(X) \neq \sigma X_i\}.$$

The stability theorem gives

$$\ln 2 - h(p) - I(f(X); Y) \geq \frac{p\rho^2}{2000} \delta(f).$$

At fixed nontrivial noise, nearly optimal information therefore implies closeness to a coordinate, independently of the number of inputs. The proof gives better piecewise coefficients; page 4 makes one use concrete. The constants are conservative.

Why the methods are worth examining separately

The first mechanism converts a scalar inequality into a bound on every Boolean cube. The second tells us which entropy quantity has a useful symmetry. The third converts a spectral cap and a scalar entropy envelope into information. Each is stated with its own inputs and conclusion, so a reader need not reuse the entire CK argument to use one part.

Reusable statement	Location in the full manuscript
Restriction–mixing criterion	Proposition 2.3, p. 6
Entropy-product centering	Lemma 3.1, p. 8
Spectral-cap / entropy-envelope certificate	Proposition 5.2, p. 17
Information-to-coordinate recovery	Corollary 7.1, p. 23

2. Low noise: identify the exact loss, then recover it

Put $\mu = \mathbb{E}f$ and $V(\mu) = 4\mu(1 - \mu)$. Fixing one coordinate produces two smaller Boolean functions f_0, f_1 , with means μ_0, μ_1 . Their fixed mean gap is $d = |\mu_1 - \mu_0|$. The law of total variance gives

$$\frac{V(\mu_0) + V(\mu_1)}{2} = V(\mu) - d^2.$$

Thus an induction with coefficient c loses exactly cd^2 . Restricting to balanced functions would not work: the two-bit AND function has a constant child and a coordinate child.

After observing the other noisy coordinates, let A, B be the two section posteriors. Bayes' formula gives

$$\mathbf{H}(f \mid Y) = \mathbb{E}M_p(A, B), \quad M_p(a, b) = \frac{h((1-p)a + pb) + h(pa + (1-p)b)}{2},$$

while $\mathbb{E}|A - B| \geq d$. The two posteriors use the *same* observation; they are not assumed independent.

The reusable implication. If $0 \leq c \leq h(p)$ and

$$(1 + d^2)M_p(a, b) \geq \frac{h(a) + h(b)}{2} + 2cd|a - b|$$

for every pair a, b and every independent $d \in [0, 1/2]$, then

$$\begin{aligned} (1 + d^2)\mathbf{H}(f \mid Y) &\geq c[V(\mu) - d^2] + 2cd\mathbb{E}|A - B| \\ &\geq c[V(\mu) + d^2] \geq (1 + d^2)cV(\mu). \end{aligned}$$

Constants and coordinates supply the base cases. A coordinate with $d \leq 1/2$ always exists when $n \geq 2$, because any two coordinate correlations have absolute values summing to at most one.

Why the scalar problem centers a product

Set $J = [h(a) + h(b)]/2$, $M = M_p(a, b)$, $K = M - J$, and $\ell = c|a - b|$. The required gap is

$$K + Md^2 - 2\ell d = M \left(d - \frac{\ell}{M} \right)^2 + \frac{KM - \ell^2}{M}.$$

This identifies KM , not M alone, as the useful object. At fixed $t = |a - b|$, the manuscript proves $K \geq K_0$ and $KM \geq K_0M_0$, where the subscript denotes the symmetric pair $((1-t)/2, (1+t)/2)$. The proof passes through an exact nonnegative curvature integral. It does *not* assert that every fixed- d gap is minimized at symmetry.

Where to slow down. Product centering and the three centered scalar regimes are the analytic work, not the induction arithmetic. For $p \leq 1/20$, they establish $c = h(p)$ and an additional surplus $(p/500)(a - b)^2$. The main manuscript's Section 3 and Appendix A prove those obligations, including deterministic boundaries.

3. Complementary noise: choose the statistic from its spectrum

For $F = 2f - 1$, let $m = \mathbb{E}F$ and $W_k(F)$ be its squared degree- k Fourier mass. Large $|m|$ or a large coordinate correlation admits a direct bound. In the remaining class, the manuscript proves

$$m^2 + W_1(F) \leq \frac{31}{40}.$$

This uses an antipodal identity, an explicit polynomial moment estimate, and a balancing lift. It does not import a specialized Fourier theorem or assume that the output is balanced.

Put $G = T_\rho F$ and $R = |G|$. The channel gives $H(f \mid Y) = \mathbb{E}\eta(R)$, where $\eta(t) = h((1+t)/2)$. Now subtract two spectral quantities:

$$\mathbb{E}G^2 - \rho^2 \mathbb{E}(FG) = \sum_{k \geq 0} (\rho^{2k} - \rho^{k+2}) W_k(F).$$

Degree two cancels. All higher degrees have a favorable sign. The product FG is evaluated at the *same uniform cube point*, not at opposite channel endpoints.

Rearranging dictates

$$\psi_\rho(t) = (1-t)(1+t-\rho^2).$$

For a general cap $m^2 + W_1(F) \leq \Omega \leq 1$, put $\Lambda_\Omega = 1 + \rho - \Omega\rho^2$ and $\Gamma = 1 + \rho - \rho^2$. The exact energy identity gives

$$\mathbb{E}\psi_\rho(R) \geq (1-\rho)(\Lambda_\Omega - \Gamma m^2).$$

The reusable certificate. Suppose $\lambda, \zeta \geq 0$ and $\eta(t) \geq \lambda\psi_\rho(t) + \zeta(1-t)$ for every $t \in [0, 1]$. Then

$$\begin{aligned} \ln 2 - \eta(\rho) - I(f; Y) &\geq \lambda(1-\rho)\Lambda_\Omega - \eta(\rho) \\ &\quad + m^2 \left[\frac{1}{2} - \lambda(1-\rho)\Gamma \right] + \zeta \mathbb{E}(1-R). \end{aligned}$$

The first term pays for the balanced target, the second accounts for output bias, and the third retains strictness. Improving the cap or the envelope is a well-defined independent task.

Where to slow down. A scalar envelope must hold on the entire interval, not at sample points. Appendix D uses convexity, logarithmic concavity, and positive Bernstein coefficients. For still higher noise, a strict anchor is required: merely contracting the coordinate bound at an intermediate correlation would give an upper bound larger than the target.

4. How to apply the stability conclusion

For $0 < p < 1/2$, the proof actually gives

$$\mathcal{D}_p(f) := \ln 2 - h(p) - \mathbb{I}(f; Y) \geq c_p \delta(f), \quad c_p = \begin{cases} 9p(1-2p)^2/10000, & 0 < p < 1/20, \\ (1-2p)^2/2000, & 1/20 \leq p < 1/2. \end{cases}$$

If a valid estimate gives $\mathcal{D}_p(f) \leq \epsilon$, choose the coordinate maximizing $|\mathbb{E}(FX_i)|$ and choose its orientation accordingly. Its disagreement probability is

$$\delta(f) = \frac{1 - \max_i |\mathbb{E}(FX_i)|}{2} \leq \min\{1/2, \epsilon/c_p\}.$$

At $p = 1/4$, $c_p = 1/8000$: a gap at most 10^{-6} nats gives disagreement at most $1/125$, in any finite dimension. This is a mathematical recovery statement, not an efficient learning algorithm or a method for estimating mutual information from data.

The converse is elementary. For a nearest coordinate indicator g , the chain rule and the error bit $f \oplus g$ give

$$\mathbb{I}(g; Y) - \mathbb{I}(f; Y) \leq H(g | f) \leq h(\delta(f)).$$

Hence $c_p \delta(f) \leq \mathcal{D}_p(f) \leq h(\delta(f))$. The rates do not match, but near optimality and closeness are equivalent qualitatively at fixed nontrivial noise.

Examples that prevent common misreadings

Parity. For $F = \prod_{i=1}^k X_i$, the exact conditional entropy is $\eta(\rho^k)$. Two-bit parity has $d = 0$ but nonzero posterior separation. Its deficit comes entirely from scalar mixing, not from the triangle gap.

AND. Its two child means are 0 and $1/2$. The general-mean induction is essential even for this elementary function. The four posterior probabilities are $p^2, p(1-p), p(1-p), (1-p)^2$.

The scalar limit. Any uniform coefficient in $S_{p,d}(a, b) \geq \gamma(p)(a-b)^2$ obeys $\gamma(p) \leq 2p(1-p)$: take centered posteriors with separation tending to zero and $d = 0$. A logarithmic small-noise deficit for a fixed Boolean function therefore cannot simply be inserted as a uniform coefficient for every posterior pair.

What is supplied for reuse

The full manuscript contains the proofs and rational constant bounds. The source package includes an equation-level change map, separately scoped regression checks, and `tools/finite_instance.py`, which computes exact finite-channel and Fourier quantities and rational entropy enclosures. It is a finite-example tool, not a substitute for the analytic proof. All manuscript locations above refer to v11; the earlier reviewed versions are preserved separately.